

$$3d^7$$

August 25, 2026

$$1 \quad \left| 3d_{xz}^2 3d_{xy}^2 3d_{yz}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle$$

Here are the CSFs:

$$\begin{aligned}
|1, 3/2, 3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|2, 3/2, 3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|3, 3/2, 3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|4, 3/2, 3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|5, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|6, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|7, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|8, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|9, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|10, 3/2, 3/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|11, 3/2, 1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

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$$\begin{aligned}
|21, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|22, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|23, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|24, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|25, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|26, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|27, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|28, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|29, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|30, 3/2, -1/2\rangle &= \frac{1}{\sqrt{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
&\quad + \frac{1}{\sqrt{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|31, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|32, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|33, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|34, 3/2, -3/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|35, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle \\
|36, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|37, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|38, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|39, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|40, 3/2, -3/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|41, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|42, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|43, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|44, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|45, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|46, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|47, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|48, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|49, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|50, 1/2, 1/2\rangle &= +\frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|51, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|52, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|53, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|54, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|55, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
|56, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|57, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|58, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|59, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|60, 1/2, -1/2\rangle &= \frac{1}{\sqrt{2}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{2}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|61, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|62, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|63, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|64, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|65, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\alpha \right\rangle \\
|66, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|67, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|68, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|69, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|70, 1/2, 1/2\rangle &= \sqrt{\frac{2}{3}} \left| 3d_{xz}^\alpha 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|71, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle
\end{aligned}$$

$$\begin{aligned}
|72, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|73, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|74, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|75, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^\alpha \right\rangle \\
|76, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|77, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|78, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|79, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|80, 1/2, -1/2\rangle &= \frac{1}{\sqrt{6}} \left| 3d_{xz}^\alpha 3d_{yz}^\beta 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle + \frac{1}{\sqrt{6}} \left| 3d_{xz}^\beta 3d_{yz}^\alpha 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
&\quad - \sqrt{\frac{2}{3}} \left| 3d_{xz}^\beta 3d_{yz}^\beta 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|81, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\
|82, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\
|83, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|84, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|85, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|86, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|87, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|88, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|89, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|90, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|91, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|92, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|93, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|94, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|95, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|96, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|97, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\
|98, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\
|99, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|100, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|101, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle
\end{aligned}$$

$$\begin{aligned}
|102, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \\
|103, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|104, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|105, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|106, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|107, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|108, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|109, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|110, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|111, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|112, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|113, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^0 \right\rangle \\
|114, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^0 3d_{x^2-y^2}^2 \right\rangle \\
|115, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^0 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|116, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|117, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\
|118, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\
|119, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\
|120, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle
\end{aligned}$$

The Ground State CSFs are 2 12 22 32

$$\begin{aligned}
\langle 2, \frac{3}{2}, \frac{3}{2} | L_y | 1, \frac{3}{2}, \frac{3}{2} \rangle &= -I \\
\langle 2, \frac{3}{2}, \frac{3}{2} | L_y | 8, \frac{3}{2}, \frac{3}{2} \rangle &= \sqrt{3}I \\
\langle 2, \frac{3}{2}, \frac{3}{2} | L_y | 9, \frac{3}{2}, \frac{3}{2} \rangle &= I
\end{aligned}$$

$$\Delta g_{yy}$$

$$\begin{aligned}
&+ \frac{2}{3} \zeta \Delta_{2-1}^{-1} \\
&+ 2 \zeta \Delta_{2-8}^{-1} \\
&+ \frac{2}{3} \zeta \Delta_{2-9}^{-1}
\end{aligned}$$

$$\begin{aligned}
\langle 2, \frac{3}{2}, \frac{3}{2} | L_z | 4, \frac{3}{2}, \frac{3}{2} \rangle &= 2I \\
\langle 2, \frac{3}{2}, \frac{3}{2} | L_z | 5, \frac{3}{2}, \frac{3}{2} \rangle &= I
\end{aligned}$$

$$\Delta g_{zz}$$

$$\begin{aligned}
&+ \frac{8}{3} \zeta \Delta_{2-4}^{-1} \\
&+ \frac{2}{3} \zeta \Delta_{2-5}^{-1}
\end{aligned}$$

For a multiplicity of 4 we have the following:

$$D = \frac{1}{2} (\langle \frac{3}{2}, \frac{3}{2} | \frac{3}{2}, \frac{3}{2} \rangle - \langle \frac{3}{2}, \frac{1}{2} | \frac{3}{2}, \frac{1}{2} \rangle)$$

$$E = \frac{1}{\sqrt{3}} \langle \frac{3}{2}, -\frac{1}{2} | \frac{3}{2}, \frac{3}{2} \rangle$$

$$\begin{aligned}
\left\langle \frac{3}{2}, \frac{3}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{3}{2} \right\rangle &= -\frac{4}{9} \zeta^2 (\\
&+ \Delta_4^{-1} + \frac{1}{4} \Delta_5^{-1} + \frac{1}{12} \Delta_{11}^{-1} + \frac{1}{4} \Delta_{18}^{-1} + \frac{1}{12} \Delta_{19}^{-1} + \frac{1}{8} \Delta_{41}^{-1} \\
&+ \frac{3}{8} \Delta_{48}^{-1} + \frac{1}{8} \Delta_{49}^{-1} + \frac{1}{24} \Delta_{61}^{-1} + \frac{1}{8} \Delta_{68}^{-1} + \frac{1}{24} \Delta_{69}^{-1} + \frac{1}{4} \Delta_{81}^{-1} \\
&+ \frac{3}{4} \Delta_{82}^{-1} + \frac{3}{4} \Delta_{90}^{-1} + \frac{1}{4} \Delta_{91}^{-1})
\end{aligned}$$

$$\begin{aligned}
\left\langle \frac{3}{2}, \frac{1}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{1}{2} \right\rangle = & -\frac{4}{9}\zeta^2 (\\
& + \frac{1}{12}\Delta_1^{-1} + \frac{1}{4}\Delta_8^{-1} + \frac{1}{12}\Delta_9^{-1} + \frac{1}{9}\Delta_{14}^{-1} + \frac{1}{36}\Delta_{15}^{-1} + \frac{1}{9}\Delta_{21}^{-1} \\
& + \frac{1}{3}\Delta_{28}^{-1} + \frac{1}{9}\Delta_{29}^{-1} + \frac{2}{3}\Delta_{44}^{-1} + \frac{1}{6}\Delta_{45}^{-1} + \frac{1}{24}\Delta_{51}^{-1} + \frac{1}{8}\Delta_{58}^{-1} \\
& + \frac{1}{24}\Delta_{59}^{-1} + \frac{2}{9}\Delta_{64}^{-1} + \frac{1}{18}\Delta_{65}^{-1} + \frac{1}{72}\Delta_{71}^{-1} + \frac{1}{24}\Delta_{78}^{-1} + \frac{1}{72}\Delta_{79}^{-1} \\
& + \frac{1}{12}\Delta_{101}^{-1} + \frac{1}{4}\Delta_{102}^{-1} + \frac{1}{4}\Delta_{110}^{-1} + \frac{1}{12}\Delta_{111}^{-1})
\end{aligned}$$

$$\begin{aligned}
\left\langle \frac{3}{2}, -\frac{1}{2} \left| \hat{H}_{eff} \right| \frac{3}{2}, \frac{3}{2} \right\rangle = & -\frac{4}{9}\zeta^2 (\\
& - \frac{1}{3\sqrt{12}}\Delta_{11}^{-1} - \frac{1}{\sqrt{12}}\Delta_{18}^{-1} - \frac{1}{3\sqrt{12}}\Delta_{19}^{-1} + \frac{\sqrt{3}}{24}\Delta_{41}^{-1} + \frac{\sqrt{3}}{8}\Delta_{48}^{-1} \\
& + \frac{\sqrt{3}}{24}\Delta_{49}^{-1} + \frac{\sqrt{3}}{72}\Delta_{61}^{-1} + \frac{\sqrt{3}}{24}\Delta_{68}^{-1} + \frac{\sqrt{3}}{72}\Delta_{69}^{-1} - \frac{\sqrt{3}}{12}\Delta_{81}^{-1} \\
& - \frac{\sqrt{3}}{4}\Delta_{82}^{-1} - \frac{\sqrt{3}}{4}\Delta_{90}^{-1} - \frac{\sqrt{3}}{12}\Delta_{91}^{-1})
\end{aligned}$$

$$\begin{aligned}
D = & \frac{2}{9}\zeta^2 (\\
& + \frac{1}{12}\Delta_1^{-1} - \Delta_4^{-1} - \frac{1}{4}\Delta_5^{-1} + \frac{1}{4}\Delta_8^{-1} + \frac{1}{12}\Delta_9^{-1} - \frac{1}{12}\Delta_{11}^{-1} \\
& + \frac{1}{9}\Delta_{14}^{-1} + \frac{1}{36}\Delta_{15}^{-1} - \frac{1}{4}\Delta_{18}^{-1} - \frac{1}{12}\Delta_{19}^{-1} + \frac{1}{9}\Delta_{21}^{-1} + \frac{1}{3}\Delta_{28}^{-1} \\
& + \frac{1}{9}\Delta_{29}^{-1} - \frac{1}{8}\Delta_{41}^{-1} + \frac{2}{3}\Delta_{44}^{-1} + \frac{1}{6}\Delta_{45}^{-1} - \frac{3}{8}\Delta_{48}^{-1} - \frac{1}{8}\Delta_{49}^{-1} \\
& + \frac{1}{24}\Delta_{51}^{-1} + \frac{1}{8}\Delta_{58}^{-1} + \frac{1}{24}\Delta_{59}^{-1} - \frac{1}{24}\Delta_{61}^{-1} + \frac{2}{9}\Delta_{64}^{-1} + \frac{1}{18}\Delta_{65}^{-1} \\
& - \frac{1}{8}\Delta_{68}^{-1} - \frac{1}{24}\Delta_{69}^{-1} + \frac{1}{72}\Delta_{71}^{-1} + \frac{1}{24}\Delta_{78}^{-1} + \frac{1}{72}\Delta_{79}^{-1} - \frac{1}{4}\Delta_{81}^{-1} \\
& - \frac{3}{4}\Delta_{82}^{-1} - \frac{3}{4}\Delta_{90}^{-1} - \frac{1}{4}\Delta_{91}^{-1} + \frac{1}{12}\Delta_{101}^{-1} + \frac{1}{4}\Delta_{102}^{-1} + \frac{1}{4}\Delta_{110}^{-1} \\
& + \frac{1}{12}\Delta_{111}^{-1})
\end{aligned}$$

$$\begin{aligned}
E = & \frac{4}{9\sqrt{3}}\zeta^2 (\\
& + \frac{1}{3\sqrt{12}}\Delta_{11}^{-1} + \frac{1}{\sqrt{12}}\Delta_{18}^{-1} + \frac{1}{3\sqrt{12}}\Delta_{19}^{-1} - \frac{\sqrt{3}}{24}\Delta_{41}^{-1} - \frac{\sqrt{3}}{8}\Delta_{48}^{-1} \\
& - \frac{\sqrt{3}}{24}\Delta_{49}^{-1} - \frac{\sqrt{3}}{72}\Delta_{61}^{-1} - \frac{\sqrt{3}}{24}\Delta_{68}^{-1} - \frac{\sqrt{3}}{72}\Delta_{69}^{-1} + \frac{\sqrt{3}}{12}\Delta_{81}^{-1} \\
& + \frac{\sqrt{3}}{4}\Delta_{82}^{-1} + \frac{\sqrt{3}}{4}\Delta_{90}^{-1} + \frac{\sqrt{3}}{12}\Delta_{91}^{-1})
\end{aligned}$$